FAN-HHTT
Electrical System

Single Line Diagram (SLD)
Project Overview & Equipment Summary

Document Field	Details
Project	FAN-HHTT Electrical System
Document Type	Single Line Diagram (SLD)
Area	Top Deck, Switchgear Room 04BFA
Switchgear Panel	04BFA

Document content arranged across 10 pages for readability. This overview is based solely on the supplied text; it is not a substitute for the approved electrical drawing.

1. Document Overview

This document describes the low-voltage (LV) switchgear arrangement for feeders 04BFA-01 through 04BFA-08 within the FAN-HHTT Electrical System project.

Document identification

Attribute	Value
Project	FAN-HHTT Electrical System
Document type	Single Line Diagram (SLD)
Location / area	Top Deck, Switchgear Room 04BFA
Equipment designation	Switchgear Panel 04BFA

Purpose

The supplied description summarizes the LV switchgear lineup, incoming supply, bus coupler configuration, protective relays, and short-circuit withstand rating. Equipment ratings and design references are retained as provided.

Scope note

The source text names feeders 04BFA-01 through 04BFA-08, while the detailed equipment list specifies MCC feeders 04BFA-01 through 04BFA-04 only. Details for feeders 05–08 were not supplied and have not been inferred.

2. System Description

The SLD represents the LV switchgear arrangement associated with panel 04BFA. The incoming supply is identified as originating from transformer TR-02, rated 400 V and 50 Hz on the LV side.

Key system elements

Element	Supplied information
LV switchgear	Panel 04BFA
Incoming source	Transformer TR-02; 400 V, 50 Hz
Bus arrangement	Two incomers and a bus coupler
Bus coupler status	Normally Open (NO)
Protection functions	50/51 and 50N/51N
Short-circuit rating	50 kA for 1 second

Interpretation

The listed configuration indicates a switchgear system with two 1600 A incomers and a 1600 A normally open bus coupler. The exact operating philosophy, interlocking, and bus-section arrangement must be confirmed against the approved SLD.

3. Incoming Supply & Transformer

Transformer TR-02 is identified as the source transformer for the LV supply.

Parameter	Value
Transformer tag	TR-02
Rated power	1 MVA
Voltage ratio	11 kV / 0.4 kV
Vector group	Dyn11
LV nominal voltage	400 V
Frequency	50 Hz

Supply path (textual summary)

Upstream HV interface → Transformer TR-02 (11 kV / 0.4 kV) → LV switchgear panel 04BFA.

Verification note

The upstream HV interface should be checked against drawing RAN-HHTT-0000EL-SLD-0018. The source text does not provide transformer impedance, earthing details, cable information, or the detailed HV protection arrangement.

4. Incoming Circuit Breakers

Two incoming circuit breakers are listed for switchgear panel 04BFA.

Device	Rating / type	Protection
Incomer A	1600 A ACB	Micrologic protection
Incomer B	1600 A ACB	Micrologic protection

Equipment notes

• ACB means Air Circuit Breaker.
• Both incomers are specified at 1600 A.
• The revision history records an update to the incomer ACB model in Rev 04.
• The exact manufacturer, model number, trip settings, and interlocking logic are not included in the supplied text.

Design checks

Confirm the approved breaker model and settings, coordination with downstream MCCBs, short-circuit breaking capacity, and any incomer/bus-coupler interlocks from the controlled engineering documents.

5. Bus Coupler & Bus Arrangement

The switchgear includes a 1600 A bus coupler ACB specified as Normally Open (NO).

Parameter	Supplied value
Device	Bus coupler
Breaker type	ACB
Rated current	1600 A
Normal status	Normally Open (NO)

Functional summary

A bus coupler can connect two bus sections when the approved operating conditions permit. The stated normally open status describes its normal configuration; it does not by itself define the transfer sequence or permitted switching conditions.

Important verification

Check the approved SLD and switching philosophy for bus-section boundaries, electrical/mechanical interlocks, source paralleling prevention, and the conditions under which the coupler may be closed.

6. Protection & Short-Circuit Rating

The source description identifies protection functions 50/51 and 50N/51N and a rated short-circuit capacity of 50 kA for 1 second.

ANSI device number	Common function
50	Instantaneous overcurrent protection
51	Time-delayed overcurrent protection
50N	Instantaneous earth/neutral overcurrent protection
51N	Time-delayed earth/neutral overcurrent protection

Short-circuit withstand

Rated value provided: 50 kA for 1 second. The relevant rating terminology and applicability to the complete assembly should be verified against the switchgear manufacturer's data and approved fault-level study.

Coordination note

Protective devices are stated to be coordinated in accordance with IEC 60947. Actual relay curves, pickup values, time delays, and discrimination results were not included in the source text.

7. MCC Feeder Schedule

The equipment list provides details for four motor control centre (MCC) feeders.

Feeder tag	Protective device	Rated current
04BFA-01	MCCB	250 A
04BFA-02	MCCB	250 A
04BFA-03	MCCB	160 A
04BFA-04	MCCB	125 A

Terminology

MCCB means Moulded Case Circuit Breaker. The values above are the feeder protective-device ratings supplied in the source text; they are not a substitute for checking load current, cable ampacity, fault duty, and coordination.

Scope limitation

Although the description refers to feeders 04BFA-01 through 04BFA-08, ratings for 04BFA-05 through 04BFA-08 were not provided. Those feeder details require confirmation from the complete SLD or downstream documentation.

8. Engineering Notes & References

The following notes and related document references were supplied.

Engineering notes

No.	Note
1	Protective devices coordinated as per IEC 60947.
2	Cable sizes and routes to be verified on site.
3	Refer to RAN-HHTT-0000EL-SLD-0018 for upstream HV interface.

Related documents

Document reference	Purpose / relationship
CEG-HHTT-DWG-7011	Layout reference
RAN-HHTT-0000EL-SLD-0020	Downstream MCC overview
RAN-HHTT-0000EL-SLD-0018	Upstream HV interface (referenced in note)

Field verification

Cable sizing and routing are explicitly identified as requiring on-site verification. Confirm that the referenced drawings are current and approved before using this summary for design, installation, testing, or operation.

9. Revisions, Approvals & Document Control

Revision history

Revision	Change description
Rev 03	Added MCC feeder 04BFA-04.
Rev 04	Updated incomer ACB model and fault level study results; dated 2020-04-30.

Approvals

Role	Initials / name supplied
Prepared by	GN
Checked by	DB
Approved by	RAN

Document-use statement

This 10-page PDF is a formatted expansion of the text provided by the requester. It preserves the supplied equipment data and references, adds explanatory organization, and flags information gaps without inventing missing ratings or design details. Use the approved controlled SLD and associated engineering documents as the authoritative source.

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